

Number system

→ In the Hindu - Arabic system, we have ten digits, namely: 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9.

• A number is denoted by a group of digits, called a numeral.

Face value and place value

Face value: The digit itself. It does not change according to its position.

Ex: 6842, the face value of 2 is 2.

Place value: The value of the digit, according to its position in the number.

70984 Ex: place value of 4 = $(4 \times 1) = 4$
place value of 8 = $(8 \times 10) = 80$
place value of 9 = $(9 \times 100) = 900$
place value of 0 = 0
place value of 7 = $(7 \times 10000) = 70000$

Various Types of Numbers:

1. Natural Numbers: counting numbers EX: 1, 2, 3, 4, ...
2. Whole Numbers: All counting numbers, together with 0, EX: 0, 1, 2, 3, 4, 5, ...
3. Integers: All counting numbers, zero and negatives of counting numbers.
EX: ... -3, -2, -1, 0, 1, 2, 3, ...

- set of positive integers = $\{1, 2, 3, 4, 5, \dots\}$
- set of negative integers = $\{-1, -2, -3, -4, -5, \dots\}$
- set of all non-negative integers = $\{0, 1, 2, 3, 4, 5, \dots\}$

4. Even Numbers: A counting number divisible by 2

Ex: 0, 2, 4, 8, 10, 12, 14, 16, 18, ...

5. Odd Numbers: A counting number not divisible by 2.

Ex: 1, 3, 5, 7, 9, 11, 13, ...

6. Prime Numbers: If it has exactly two factors, namely, itself and 1. All prime numbers are less than 100:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 53, 59, 61, 67,

71, 73, 79, 83, 89, 97.

7. Composite Numbers: All counting numbers which are not prime are called composite numbers. A composite number has more than 2 factors.

Ex: • 4 → factors: 1, 2, 4 → composite

• 6 → factors: 1, 2, 3, 6 → composite

• 9 → factors: 1, 3, 9 → composite

• 12 → factors: 1, 2, 3, 4, 6, 12 → composite

8. Perfect Numbers: A number, the sum of whose factors is equal to the number, is called a perfect number.

• Ex 1: The factors of 6 are 1, 2, 3 and 6. And $1+2+3=6$

• Ex 2: The factors of 28 are 1, 2, 4, 7, 14 and 28

And $1+2+4+7+14=28$

9. co-prime (or Relative Prime): Two numbers whose H.C.F is 1 are called co-prime numbers.

EX: (2,3) and (8,9) are pairs of co-prime

10. Twin prime: Two prime numbers whose difference is 2 are called twin-primes.

EX: (3,5), (5,7) and (11,13) are pairs of twin-prime.

11. Rational Numbers: Numbers can be expressed in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$, are called rational numbers.

EX: $\frac{1}{8}$, $-\frac{8}{11}$, 0, 6, $5\frac{2}{3}$, etc.

12. Irrational Numbers: Numbers which when expressed in decimal form are non-terminating and non-repeating are called irrational numbers.

EX: $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, $\sqrt{7}$, π , e , 0.23176.....

Important Facts:

1. Arithmetic operations rules:

- Even + Even = Even
- odd + odd = Even
- Even + odd = odd
- Even - Even = Even
- odd - odd = Even
- Even - odd = odd
- odd x Even = Even
- Even x Even = Even

- odd x odd = ~~odd~~ odd
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2. The smallest prime number is 2.
3. the only even prime number is 2.
4. The first odd prime number is 3.
5. 1 is a unique number - neither prime nor composite.
6. The least composite number is 4.
7. The least odd composite number is 9.

Test for a Number to be prime

let 'p' be a given number and let 'n' be the smallest counting number such that $\boxed{n^2 \geq p}$

Ex: i) Test 137: we know $(12)^2 > 137$. Prime numbers less than 12 are 2, 3, 5, 7, 11. clearly, none of them divider 137.

$\therefore$ 137 is a prime number.

ii) Test 173: we know $(14)^2 > 173$. Prime numbers less than 14 are 2, 3, 5, 7, 11, 13. clearly, none of them divider 173.

$\therefore$ 173 is a prime number.

Important Formulas:

$$(a+b)^2 = a^2 + b^2 + 2ab$$

$$(a-b)^2 = a^2 + b^2 - 2ab$$

$$(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$$

$$(a+b)^2 - (a-b)^2 = 4ab$$

$$(a+b)^3 = a^3 + b^3 + 3ab(a+b)$$

$$(a-b)^3 = a^3 - b^3 - 3ab(a-b)$$

$$a^2 - b^2 = (a+b)(a-b)$$

$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab+bc+ca)$$

$$a^3 + b^3 = (a+b)(a^2 + b^2 - ab)$$

$$a^3 - b^3 = (a-b)(a^2 + b^2 + ab)$$

$$a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\text{If } a+b+c=0, \text{ then } a^3 + b^3 + c^3 = 3abc$$

Test of Divisibility:

Divisibility By 2: A number is divisible by 2.

→ If its unit digit is any of 0, 2, 4, 6, 8.

Ex: 58694 is divisible by 2, while 86945 is not.

• Divisibility by 3: A number is divisible by 3 only when the sum of its digits is divisible by 3.

Ex: 695421: sum = 27, divisible by 3.

734519: sum = 29, not divisible by 3.

• Divisibility by 9: A number is divisible by 9 only when the sum of the digits is divisible by 9.

Ex: 246591: sum = 27, divisible by 9.

734519: sum = 29, not divisible by 9.